

Jaypee University of Engineering & Technology, Guna

T-1 (Even Semester- 2023)

18B14CI645 – GRAPH ALGORITHMS AND APPLICATIONS

Maximum Marks: 15

Maximum Duration: 1 Hour

Notes:

1. This question paper has 3 questions.
2. Write relevant answers only.
3. Do not write anything on question paper (Except your Er. No.).

Marks
[05]CO No.
CO4

- Q1. The streets of a small town are arranged as in Fig.1. Each edge represents a block on a particular street and each vertex is the intersection of where one or more blocks meet. Assume that the town's snowplow garage is located at intersection d. Prove that it is possible for the snowplow to plow each street exactly once without revisiting any block such that the snowplow returns to its garage immediately after plowing the last block. Find a route that the snowplow could take to accomplish the task (also show the graphical iteration).

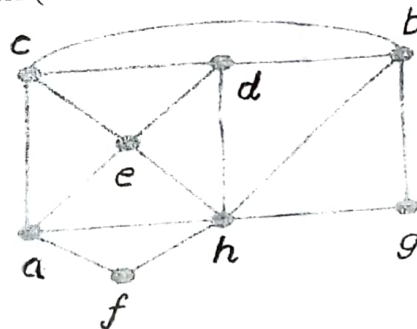


Fig.1

- Q2. (a) If G be a connected simple planar graph (n -vertices, m -edges) in which every region is bounded by at least k -edges, then show that:

[03]

CO2

$$m(k-2) \leq k(n-2).$$

- (b) A connected planar graph has nine vertices having degrees 2, 2, 2, 3, 3, 3, 4, 4 and 5. Find the number of edges and number of faces (regions) present in the graph.

[02]

CO3

- Q3. Prove that, there exists a simple graph with 12 vertices and 28 edges such that the degree of each vertex is either 3 or 5. Draw this graph.

[05]

CO5

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T-1(Even Semester 2023)**18B14CE852 - INTRODUCTION TO DISASTER MANAGEMENT**

Maximum Marks: 15

Maximum Duration: 1 Hour

Notes:

1. This question paper has 04 questions.
2. Write relevant answers only.
3. Do not write anything on question paper (Except your Er. No.).

	Marks	CO No.
Q1. Define vulnerability in context of different disasters. Explain any three vulnerabilities with their contributing factors.	[04]	CO3
Q2. What are different aims of disaster response? List various disaster response activities.	[04]	CO1
Q3. Elaborate various disaster mitigation strategies. How disaster mitigation is related to development?	[04]	CO3
Q4. Explain Hazard, Vulnerability, Risk giving suitable example and their relationship with each other.	[03]	CO2

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T-1 (Even Semester 2023)

18B14EC642-EMBEDDED SYSTEM / 18B14CI544- EMBEDDED SYSTEMS

Maximum Marks: 15

Maximum duration: 1 Hour

Notes:

1. This question paper has **five** questions.
2. Write relevant answers only.
3. Do not write anything on question paper.

		Marks	CO No.
Q1.	Demonstrate and sketch the block diagram of handheld computer embedded system.	[05]	CO3
Q2.	Draw and describe the each flag of flag register in 8085 microprocessor.	[03]	CO2
Q3.	Distinguish the difference between microprocessor and microcontroller.	[02]	CO4
Q4.	Elaborate different challenges and issues in designing Embedded systems.	[03]	CO2
Q5.	State the following pin functions of 8085 microprocessor: -	[02]	CO1
	(i) READY,		
	(ii) HOLD,		
	(iii) AD ₀ – AD ₇ ,		
	(iv) TRAP		